Exam Rules

1. 闭卷考试。
2. 可以带笔，尺等文具，计算器，如果需要可以要求考场老师提供英文词典。
3. 考场提供草稿纸，和两页（4 pages）公式纸。不允许自带草稿纸，课件，书籍，作业或者任何其它材料。公式纸会在考试前两周上传到Moodle。（公式纸包含了考试用到的大部分公式，但可能不是全部。）
4. 考试完以后草稿纸和题目纸不允许带出考场。
5. 答题纸上不允许写答案以外的任何东西（如抄写问题题目等）。



广 东 以 色 列 理 工 学 院
Guangdong Technion-Israel Institute of Technology

Lecture 12 – Polymers

Polymers

Benefits of using polymers

- Low production cost - by automatic manufacturing processes, mass production.
- Low density, high specific strength (strength/density)
- Electrical Insulation
- Thermal Insulation - handles kitchen
- High corrosion resistance

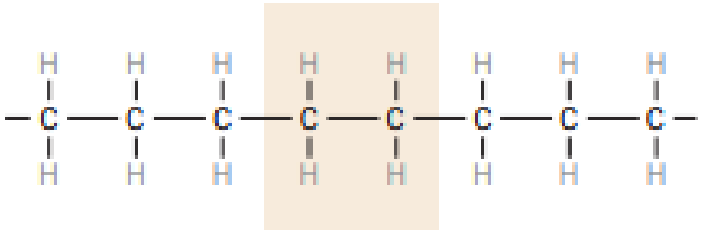
Disadvantages:

- Low durability under UV radiation
- Low durability at high temperatures

Polymers - Definitions

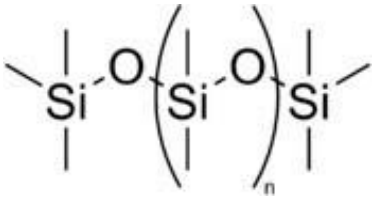
- Poly (Greek) - many, mer - Part
- Polymer - a large molecule consisting of (at least five) repeated chemical units ("mers") joined together, like beads on a string

Polyethylene



- ➢ Organic Polymers - the basic repeating units (monomers) are composed of carbon and hydrogen - Hydrocarbon Molecule.
- ➢ Inorganic Polymers - the basic repeated units are composed of silicon (oxygen and hydrogen).

Silicon Rubber



Silicon Sealant



Polymers - Definitions

- Homo-polymer - Polymer that is composed of identical monomers
- Co-polymer - Polymer that is composed of different monomers

Different types of copolymers

A , B – different monomers

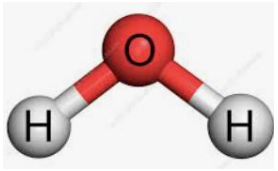
—A—B—A—B—A—B—A—B—A—B— Alternating copolymer

—A—B—B—B—A—B—A—B—A—A— Random copolymer

—B—B—B—B—B—A—A—A—A—A— Block copolymer

- The intramolecular bonds between the monomers are strong Covalent bonds
- Between the molecules, there are weak hydrogen and van der Waals bonds, and thus these polymers have relatively low melting and boiling points compared to metals and ceramics.

Less symmetry in the molecule – stronger dipoles - stronger van der Waals bond

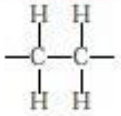
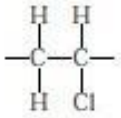
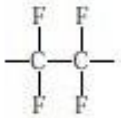
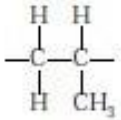
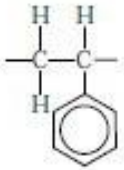
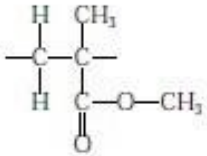
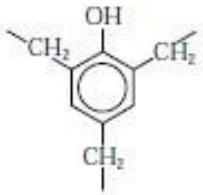
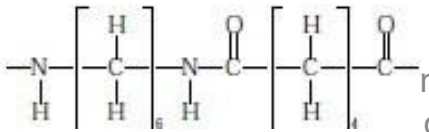


H_2O : T_m = 0 °C



CO_2 : T_m = - 44 °C

Common Polymer structures and applications

Polymer	Repeating (Mer) Structure	Uses
Polyethylene (PE)		Tubing, film, bottles, cups, electrical insulation, packaging
Polyvinyl chloride (PVC)		Architectural uses (windows frames, floor coating, artificial leather, hoses, clothing
Polytetrafluoroethylene (PTFE)		Teflon, very low friction and adhesion characteristic.
Polypropylene (PP)		Same uses as PE, but litter, stiffer, more resistance to sunlight
Polystyrene (PS)		Cheap molded objects
Polymethyl methacrylate (PMMA)		Perspex, transparent sheets, aircraft windows, laminated windscreen
Phenol-formaldehyde (Bakelite)		Handles, Formica
Polyhexamethylene adipamide (nylon 6,6)		Textiles, rope, moldings

Molecular weight of the polymer

- During the polymerization process not all polymer chains will grow to the same length; this results in a distribution of chain lengths or molecular weights
- Boiling temperatures rise with increasing molecular weight

- Number-average molecular weight:

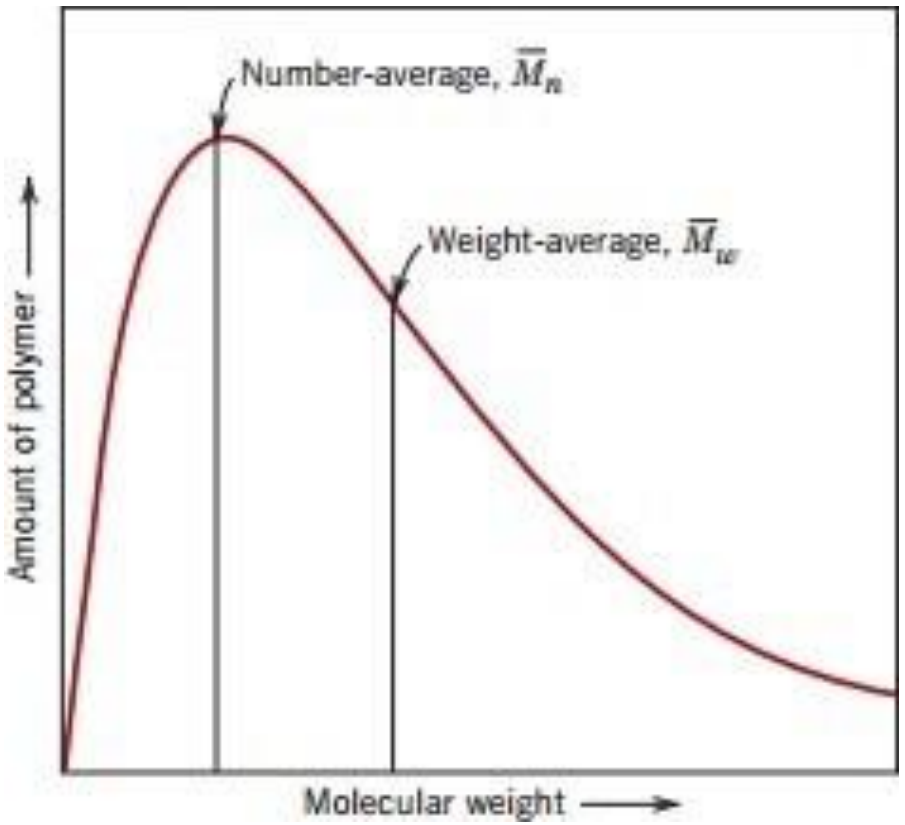
$$\overline{M}_n = \sum x_i M_i$$

M_i - Molecular weight of polymer chains with repeating unit of i.
x_i – Number fraction of M_i.

- Weight-average molecular weight:

$$\overline{M}_w = \sum w_i M_i$$

w_i - weight fraction of M_i.



Molecular Structure of Polymer

- The physical properties of polymers depend not only on molecular weight, but also on molecular structure.
- Polymers can be classified according to their molecular structure:

  - Linear polymer - repeat units (bifunctional monomer) are joined together end to end in single chains (spaghetti). Polymer chains are held together by Van der Waals bonding.

    Examples: Polyethylene, PVC, Nylon



  - Branched polymer - Side-branch chains are connected by covalent bonds to the main ones. Van der Waals bonding between chains.



Molecular Structure of Polymer

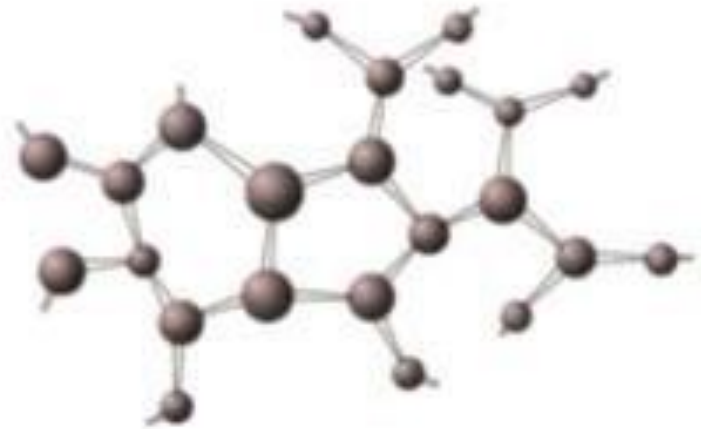
- Crosslinked polymer - adjacent linear chains are joined one to another at various positions by covalent bonds.

Examples: Rubber.



- Network polymer - Multifunctional monomers forming three or more active covalent bonds, make three-dimensional networks.

Examples: Epoxies, Phenol formaldehyde (Bakelite)



Copolymers (composed of different mers)

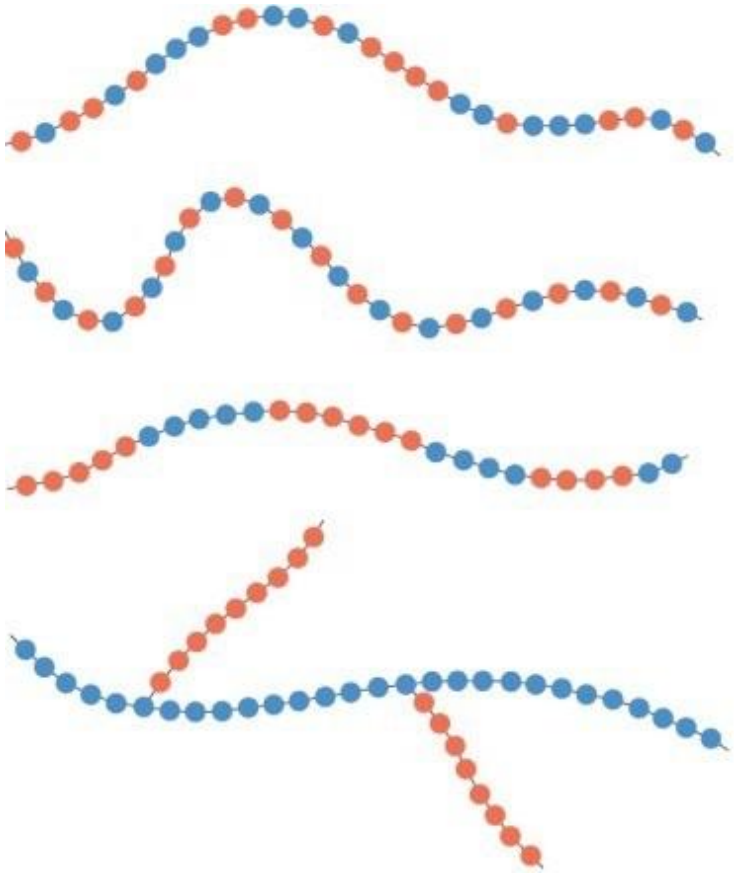
Copolymers - polymers with at least two different types of mers (repeating units), can differ in the way the mers are arranged:

Random copolymer

Alternating copolymer

Block copolymer

Graft copolymer



Thermoplastic and Thermosetting Polymers

Polymers can be classified according to their response with rising temperature:

1. Thermoplastic polymers: have linear and branched structures; they soften when heated and harden when cooled.

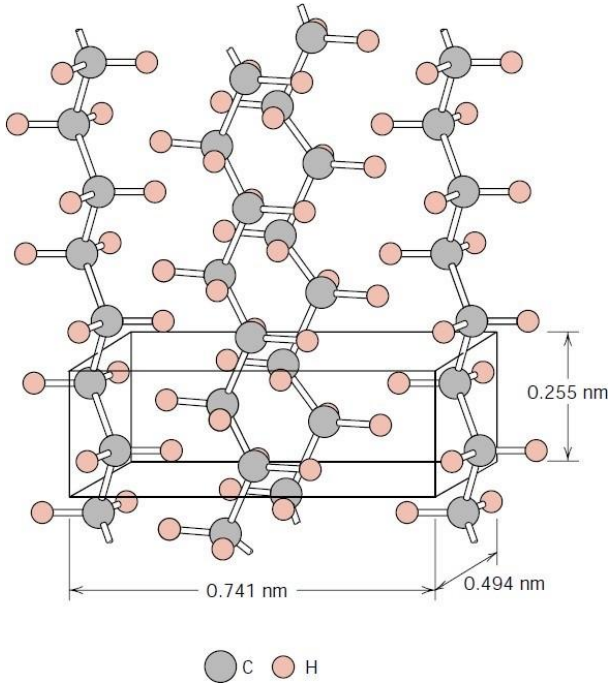
2. Thermoset polymers: once having hardened, will not soften upon heating; Their structures are crosslinked. Thermoset plastics decomposes before softening.

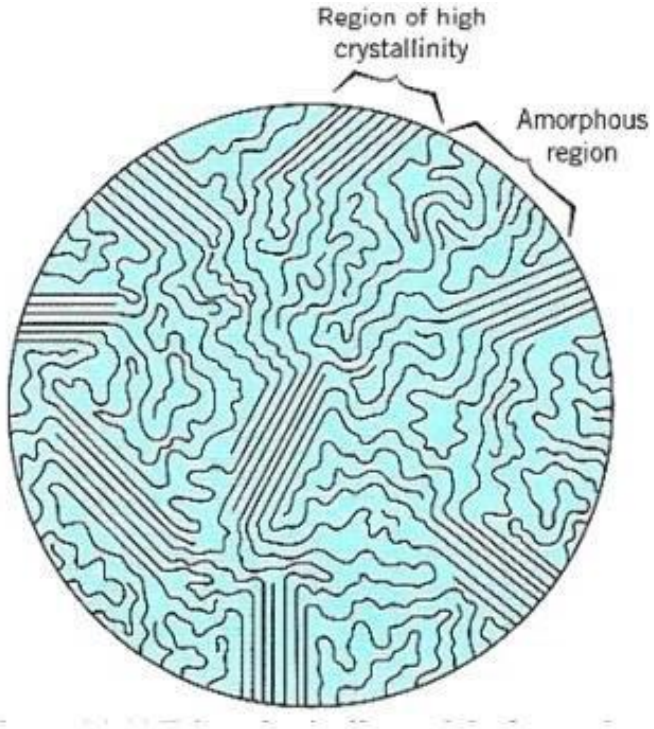
Polymer Crystallinity

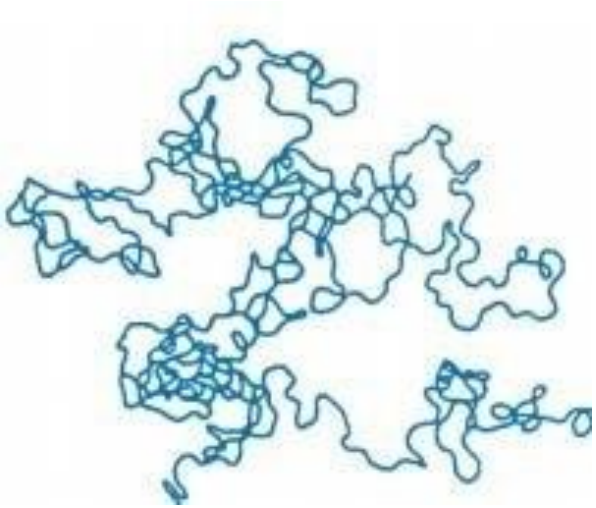
- Atomic arrangement in polymer crystals is more complex than in metals or ceramics (unit cells are typically large and complex).
- Polymer molecules are often partially crystalline (semicrystalline), with crystalline regions dispersed within amorphous material
- Polymers can be classified by their level of crystallinity:

1. Crystalline
2. Semicrystalline
3. Amorphous

Polyethylene







Polymer Crystallinity

- Crystalline polymers are denser than amorphous polymers, so the degree of crystallinity can be obtained from the measurement of density:

$$\%\, crystallinity = \frac{\rho_c(\rho_s - \rho_a)}{\rho_s(\rho_c - \rho_a)} x100$$

  - ρ_c - Density of crystalline phase
  - ρ_a - Density of amorphous phase
  - ρ_s - Density of partially crystalline sample that isanalyzed

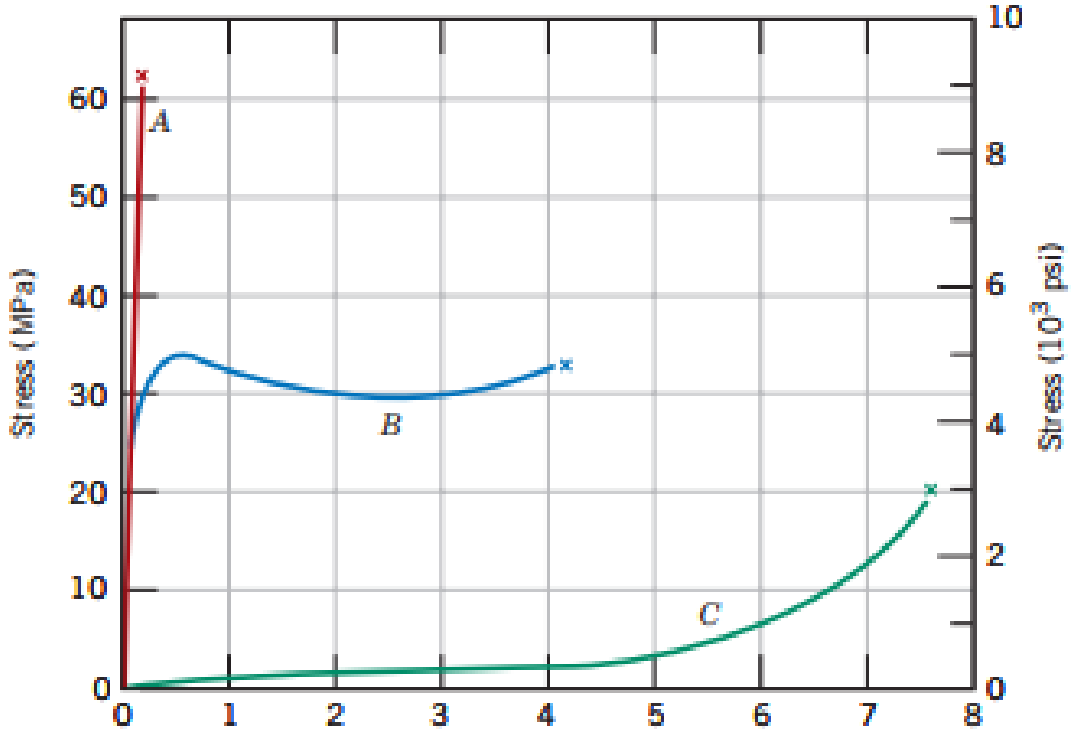
Polymer Crystallinity

- Degree of crystallinity is determined by:
  - Rate of cooling: time is necessary for chains to move and align into a crystal structure
  - Monomer complexity: crystallization less likely in complex structures. Simple polymers, such as polyethylene, crystallize relatively easily.
  - Chain configuration:
    - Linear polymers crystallize relatively easily
    - Branches inhibit crystallization
    - Network polymers almost completely amorphous
    - Crosslinked polymers can be both crystalline and amorphous
- Higher crystallinity - higher density, higher strength, higher resistance to dissolution and softening by heating

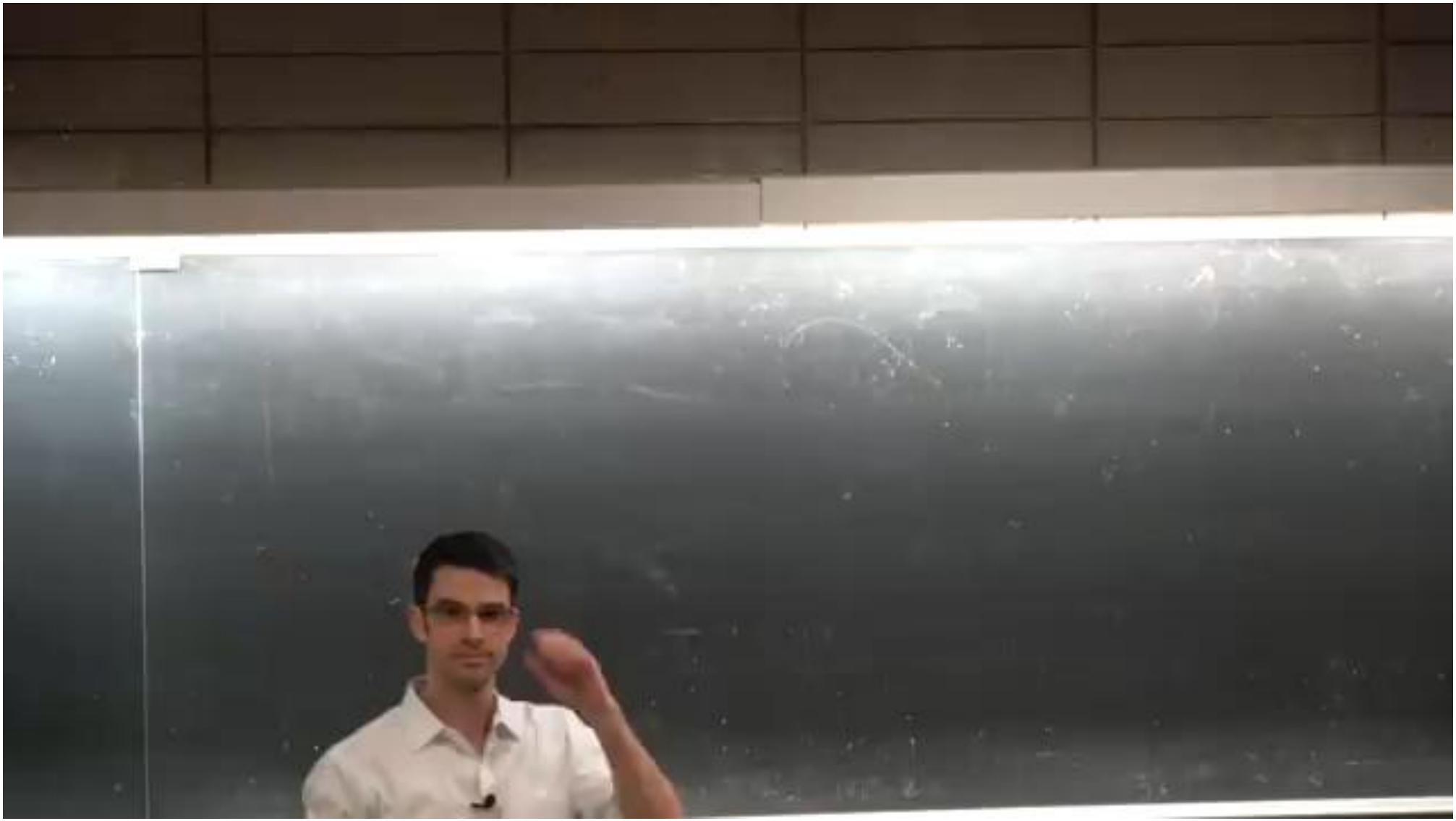
Mechanical Behavior of Polymers

Three types of stress–strain behavior for polymeric materials:

1. Curve A - Typical for brittle polymer, it fractures while deforming elastically.
2. Curve B – Typical behavior for a plastic material; is similar to that for many metallic materials.
3. Curve C – totally elastic deformation; typical for rubber which is displayed by a class of polymers termed the elastomers



Mechanical Properties of Polymers

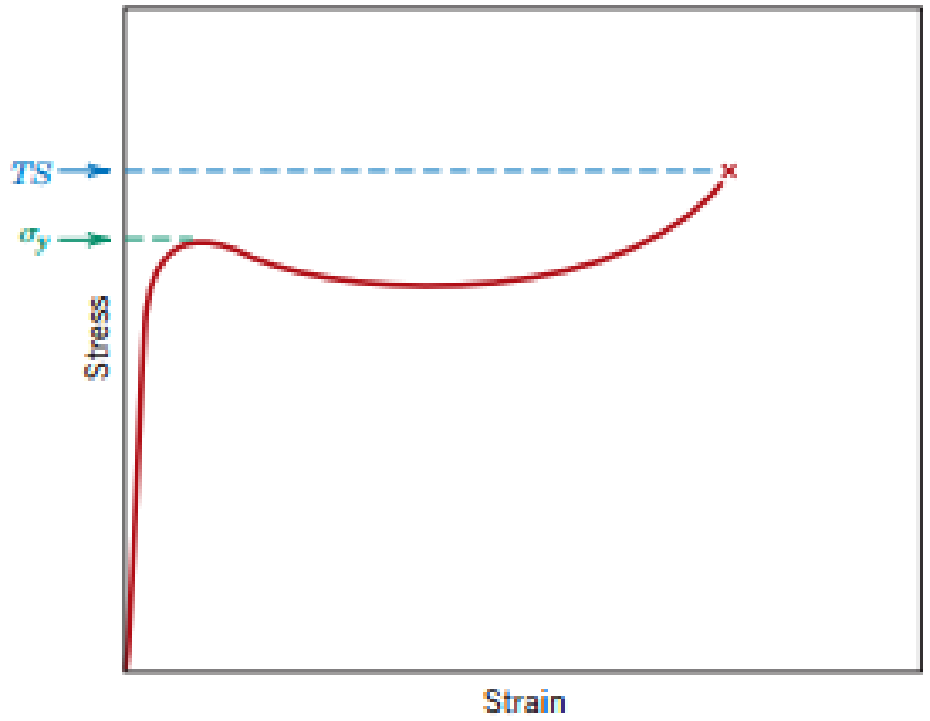


https://www.youtube.com/watch?v=vneTd1eAkE4

Stress - Strain Behavior of Polymers

Characteristics of stress-strain behavior for plastic polymers (curve B):

- Modulus of elasticity – defined as for metals
- Ductility (%EL) – defined as for metals
- Yield strength (σ_y) – is defined by the maximum on curve just beyond the elastic region (different from metals)
- Tensile strength (TS) is defined at the fracture point and can be lower than the yield strength (different from metals)



Mechanical Properties of Polymers

Table 15.1 Room-Temperature Mechanical Characteristics of Some of the More Common Polymers

Material	*Specific Gravity*	*Tensile Modulus [GPa (ksi)]*	*Tensile Strength [MPa (ksi)]*	*Yield Strength [MPa (ksi)]*	*Elongation at Break (%)*
Polyethylene (low density)	0.917–0.932	0.17–0.28 (25–41)	8.3–31.4 (1.2–4.55)	9.0–14.5 (1.3–2.1)	100–650
Polyethylene (high density)	0.952–0.965	1.06–1.09 (155–158)	22.1–31.0 (3.2–4.5)	26.2–33.1 (3.8–4.8)	10–1200
Poly(vinyl chloride)	1.30–1.58	2.4–4.1 (350–600)	40.7–51.7 (5.9–7.5)	40.7–44.8 (5.9–6.5)	40–80
Polytetrafluoroethylene	2.14–2.20	0.40–0.55 (58–80)	20.7–34.5 (3.0–5.0)	—	200–400
Polypropylene	0.90–0.91	1.14–1.55 (165–225)	31–41.4 (4.5–6.0)	31.0–37.2 (4.5–5.4)	100–600
Polystyrene	1.04–1.05	2.28–3.28 (330–475)	35.9–51.7 (5.2–7.5)	—	1.2–2.5
Poly(methyl methacrylate)	1.17–1.20	2.24–3.24 (325–470)	48.3–72.4 (7.0–10.5)	53.8–73.1 (7.8–10.6)	2.0–5.5
Phenol-formaldehyde	1.24–1.32	2.76–4.83 (400–700)	34.5–62.1 (5.0–9.0)	—	1.5–2.0
Nylon 6,6	1.13–1.15	1.58–3.80 (230–550)	75.9–94.5 (11.0–13.7)	44.8–82.8 (6.5–12)	15–300
Polyester (PET)	1.29–1.40	2.8–4.1 (400–600)	48.3–72.4 (7.0–10.5)	59.3 (8.6)	30–300
Polycarbonate	1.20	2.38 (345)	62.8–72.4 (9.1–10.5)	62.1 (9.0)	110–150

	Modulus of Elasticity	
Metal Alloy	*GPa*	*10^6 psi*
Aluminum	69	10
Brass	97	14
Copper	110	16
Magnesium	45	6.5
Nickel	207	30
Steel	207	30
Titanium	107	15.5
Tungsten	407	59

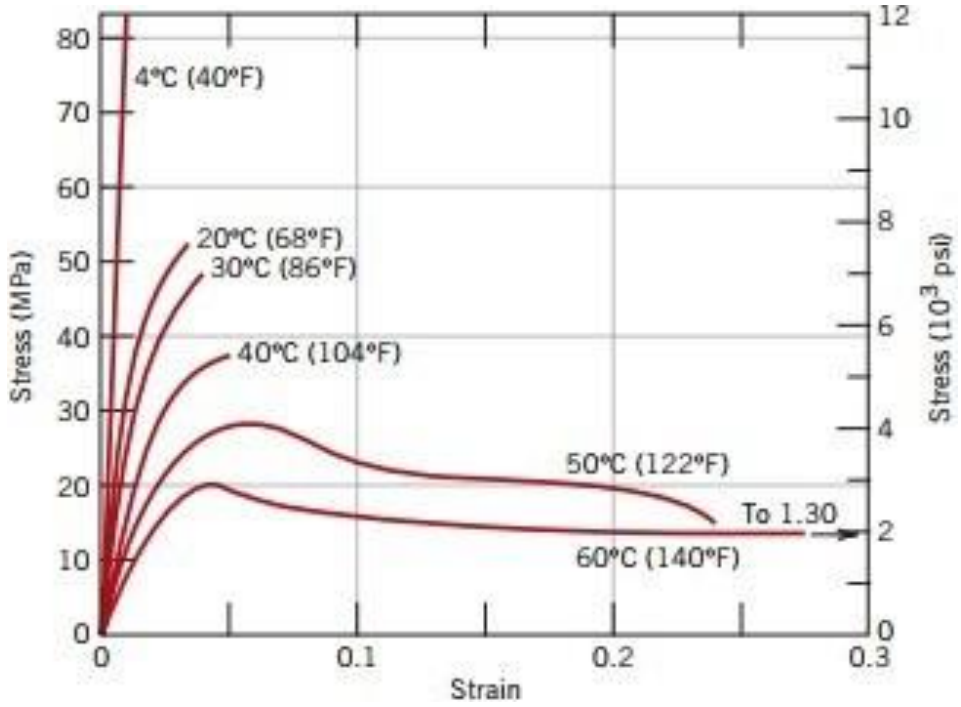
Metal Alloy	*Yield Strength MPa (ksi)*	*Tensile Strength MPa (ksi)*	*Ductility, %EL [in 50 mm (2 in.)]*
Aluminum	35 (5)	90 (13)	40
Copper	69 (10)	200 (29)	45
Brass (70Cu–30Zn)	75 (11)	300 (44)	68
Iron	130 (19)	262 (38)	45
Nickel	138 (20)	480 (70)	40
Steel (1020)	180 (26)	380 (55)	25
Titanium	450 (65)	520 (75)	25
Molybdenum	565 (82)	655 (95)	35

Mechanical Properties of Polymers

Mechanical properties of polymers change dramatically with temperature.

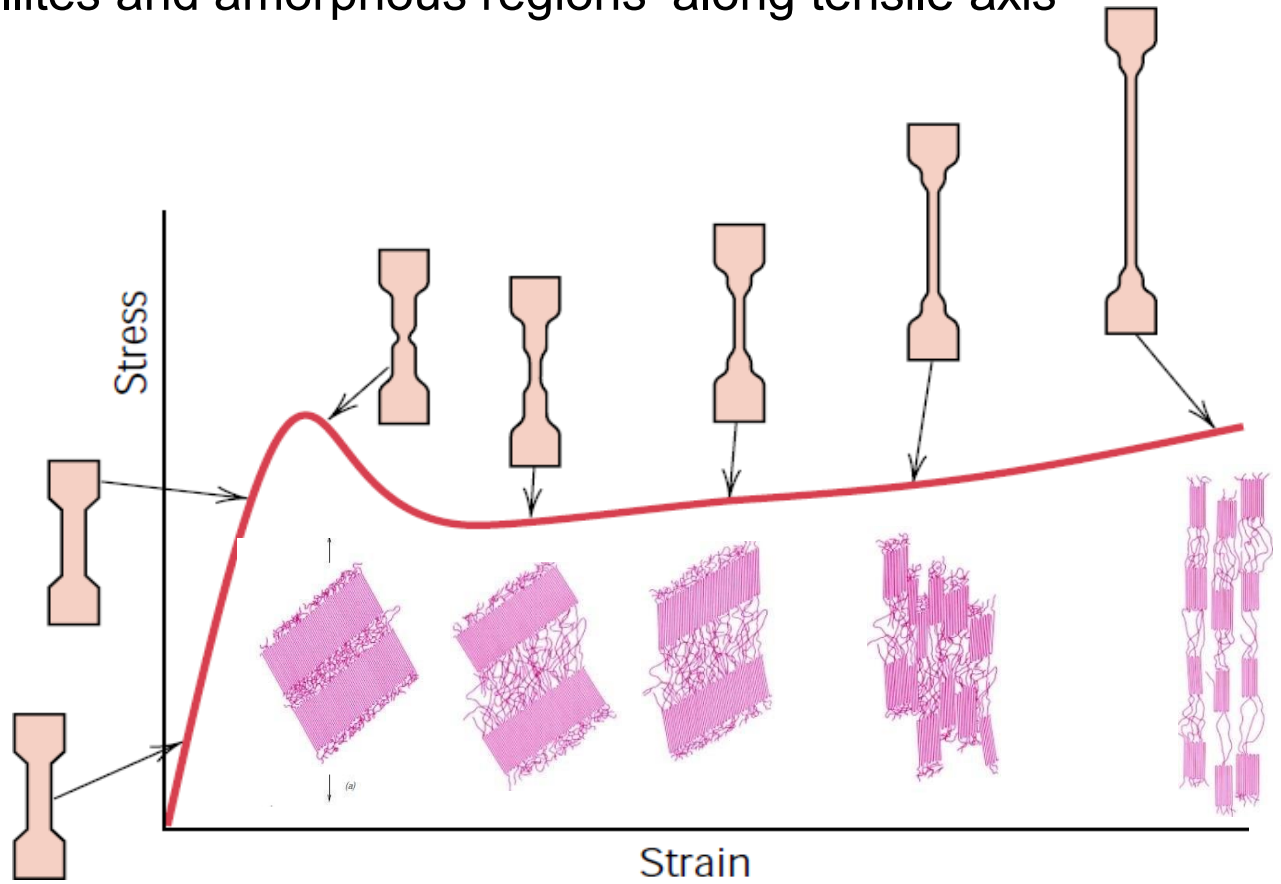
- At low temperature (~ 4°C)the material is totally brittle.
- Increasing the temperature produces:
  1. Decrease in elastic modulus
  2. Reduction in tensile strength
  3. Enhancement of ductility
- Polymers are also very sensitive to the rate of deformation (strain rate). Decreasing rate of deformation has the same effect as increasing T.

The influence of temperature on the stress–strain characteristics of **PMMA -poly(methyl methacrylate**）



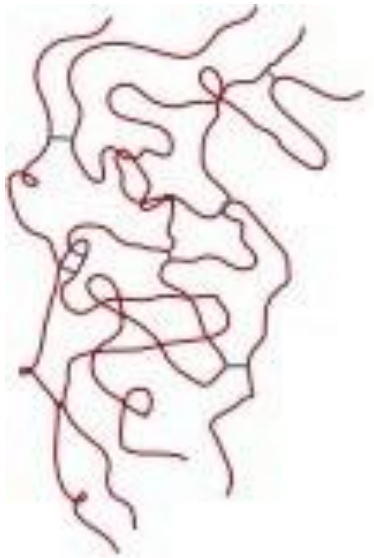
Plastic Deformation of Semicrystalline Polymers

- Plastic deformation involves the formation of neck in the specimen.
- Stages of plastic deformation:
  1. Elongation of amorphous tie chains.
  2. Tilting of lamellar crystallites towards the tensile axis
  3. Separation of crystalline block segments
  4. Stretching of crystallites and amorphous regions along tensile axis

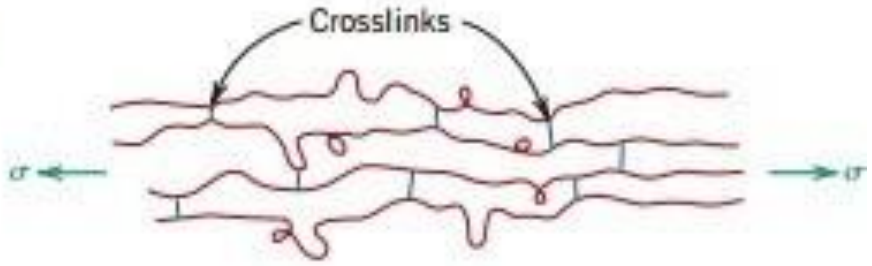


Deformation of Elastomers / Rubber

- Mechanical behavior of elastomers is characterized by large deformations and then elastically spring back to the original form.
- The moduli of elasticity is small and varies with strain since the stress– strain curve is nonlinear. This results from crosslinks in the polymer.
- In an unstressed state (a), an elastomer will be amorphous and composed of crosslinked molecular chains that are highly twisted, kinked, and coiled.
- Upon tensile load (b), the chains are partial uncoiling, untwisting, and straightening, resulting the elongation of the chains in the stress direction.



(a)

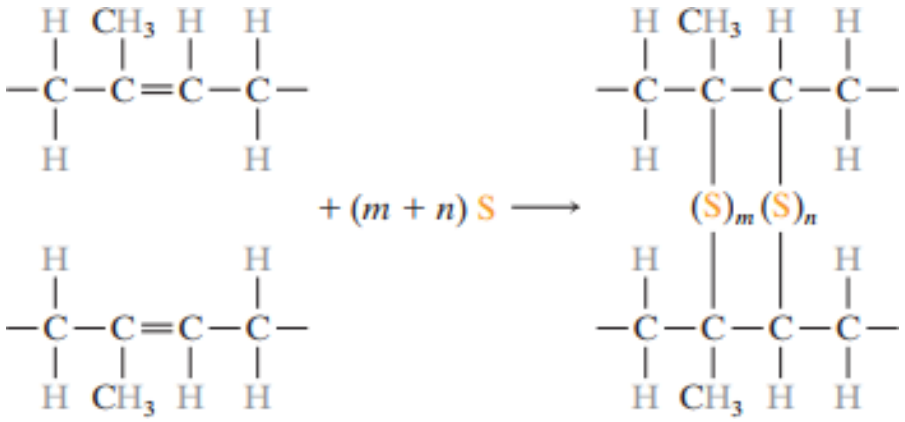


(b)

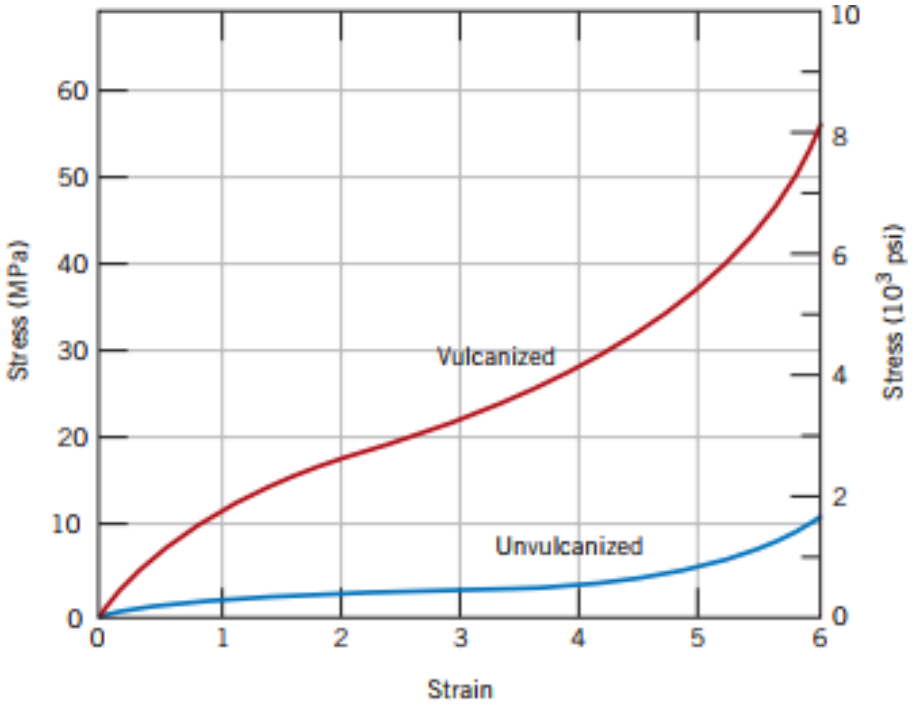
Vulcanization

- The crosslinking process in rubber is called vulcanization.
- In most vulcanizing reactions, sulfur compounds are added to the heated elastomer and crosslink them.
- Unvulcanized rubber is soft and tacky and has poor resistance to abrasion.
- Modulus, tensile strength, and resistance to degradation by oxidation are all enhanced by vulcanization.

Vulcanization process



Stress–strain curves for unvulcanized and vulcanized natural rubber



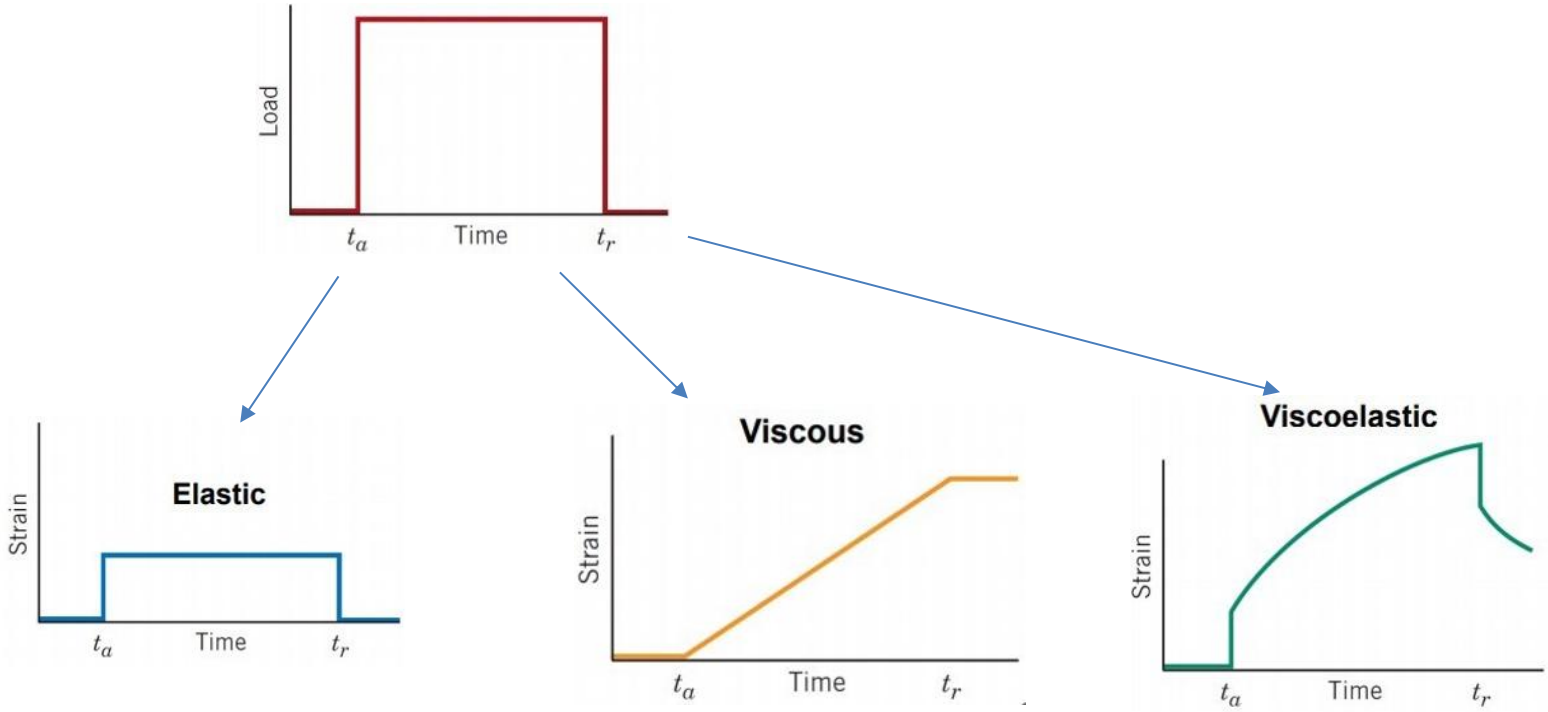
What is vulcanized Rubber (2 mins)

https://www.youtube.com/watch?v=Z_AM9ycT98k

Viscoelastic behaviour of Amorphous Polymers

The mechanical behavior of amorphous polymer is temperature-dependent:

- Low temperatures - like a glass; elastic deformation at small strains (Hooke's law, σ = Eε). Deformation is instantaneous when load is applied and it's reversible
- High temperatures: viscous or liquid behavior is dominant. Deformation is time dependent and not reversible.
- Intermediate temperatures the polymer is a rubbery solid that exhibits viscoelastic behavior that combined mechanical characteristics of both extremes.



Viscoelastic behaviour of polymers: Spring-Dash Pot model （2mins）

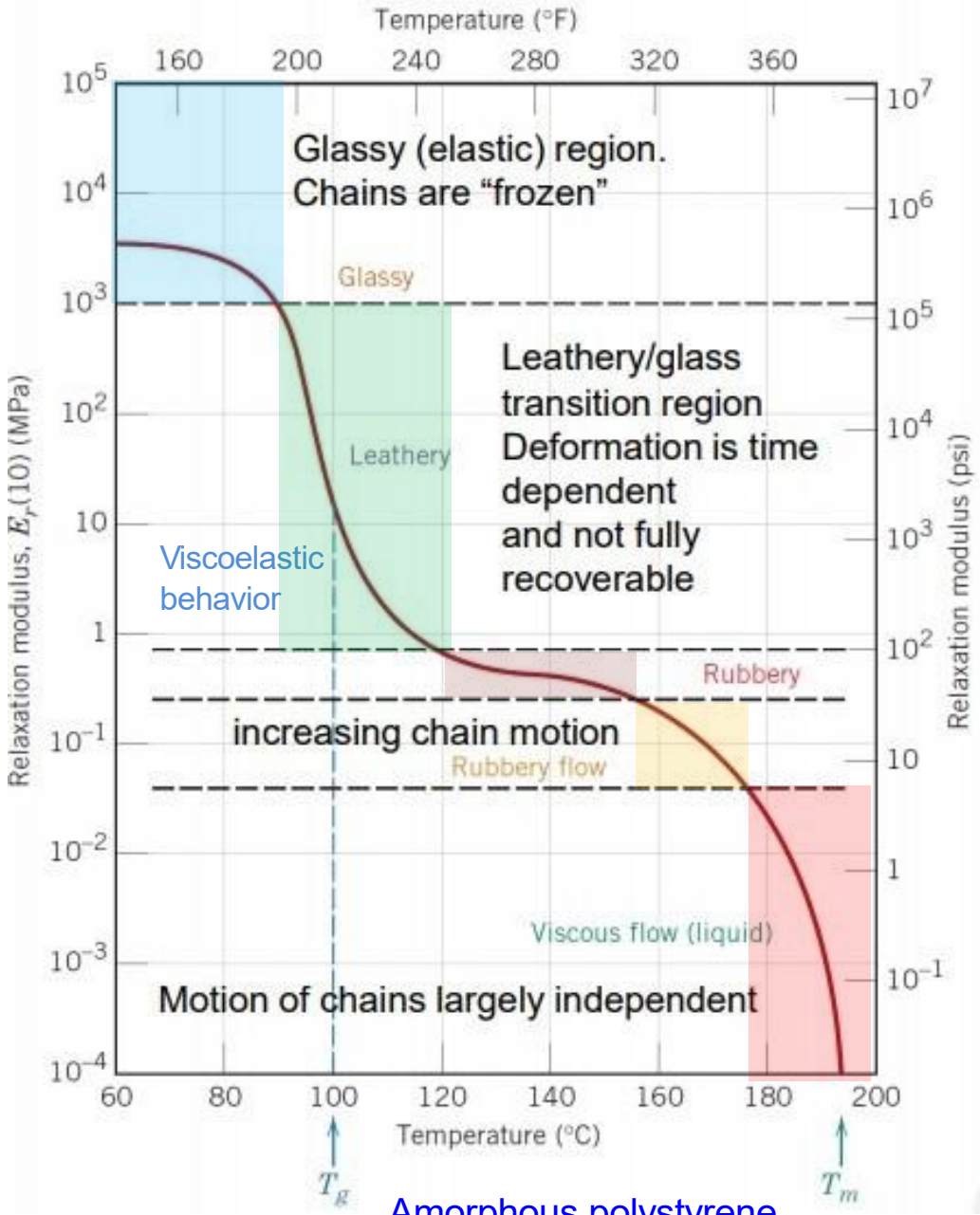


https://www.youtube.com/watch?v=q9emsMcG8cc

The influence of temperature on Modulus of Amorphous Polymers

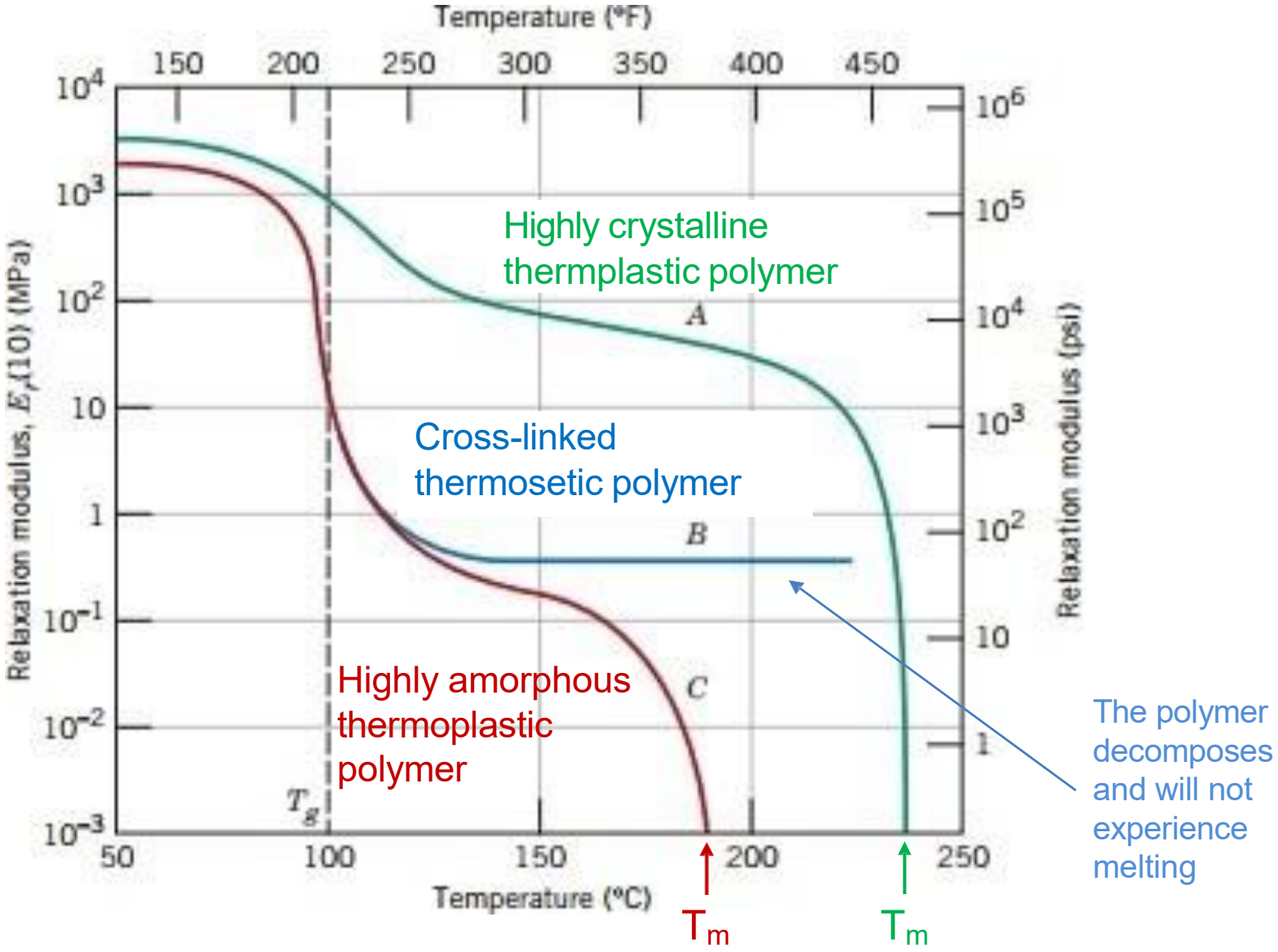
T_g - glass transition temperature

T_m – melting temperature



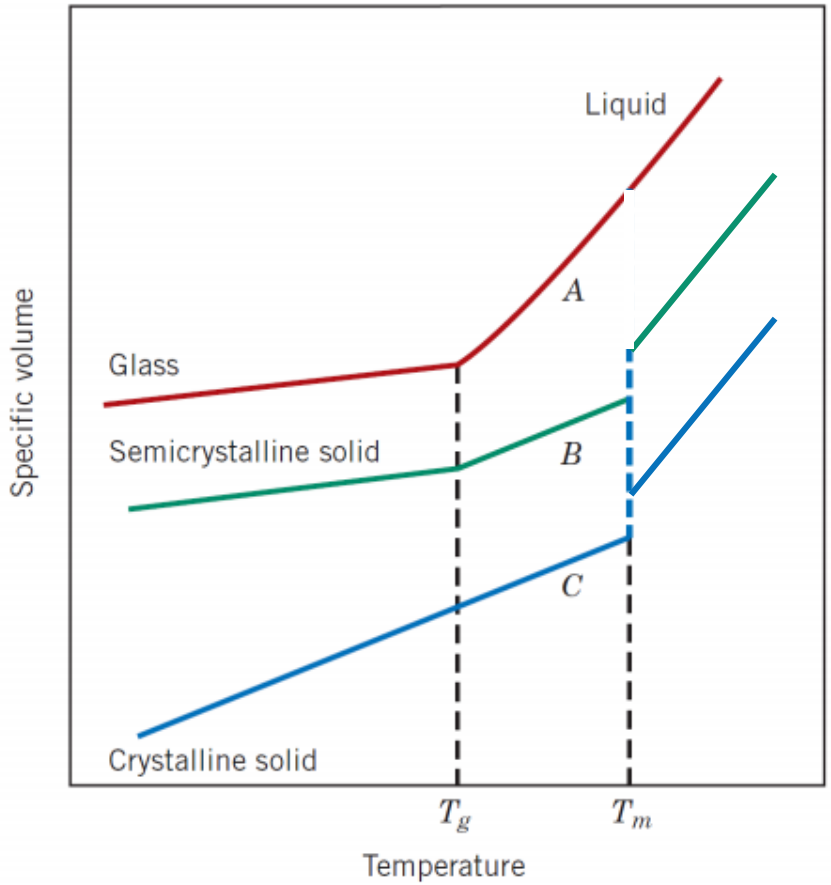
Amorphous polystyrene

Modulus vs Temperature for different polymers



Crystallization, Melting, Glass Transition

- **Glass transition:** Upon cooling, amorphous polymer/glass becomes rigid solids (reduction in specific volume), whereas the structure retains amorphous.
- Melting behavior of semicrystalline polymers is intermediate between that of crystalline materials (sharp density change at a melting temperature) and that of a pure amorphous material (melting can be only defined from viscosity).



Glass Transition & Melting temperatures

Material	*Glass Transition Temperature* [°C (°F)]	*Melting Temperature* [°C (°F)]
Polyethylene (low density)	−110 (−165)	115 (240)
Polytetrafluoroethylene	−97 (−140)	327 (620)
Polyethylene (high density)	−90 (−130)	137 (279)
Polypropylene	−18 (0)	175 (347)
Nylon 6,6	57 (135)	265 (510)
Polyester (PET)	69 (155)	265 (510)
Poly(vinyl chloride)	87 (190)	212 (415)
Polystyrene	100 (212)	240 (465)
Polycarbonate	150 (300)	265 (510)

Glass transition temperature of copolymer:

$$\frac{1}{T_g} = \frac{x_{W_1}}{T_{g1}} + \frac{x_{W_2}}{T_{g2}}$$

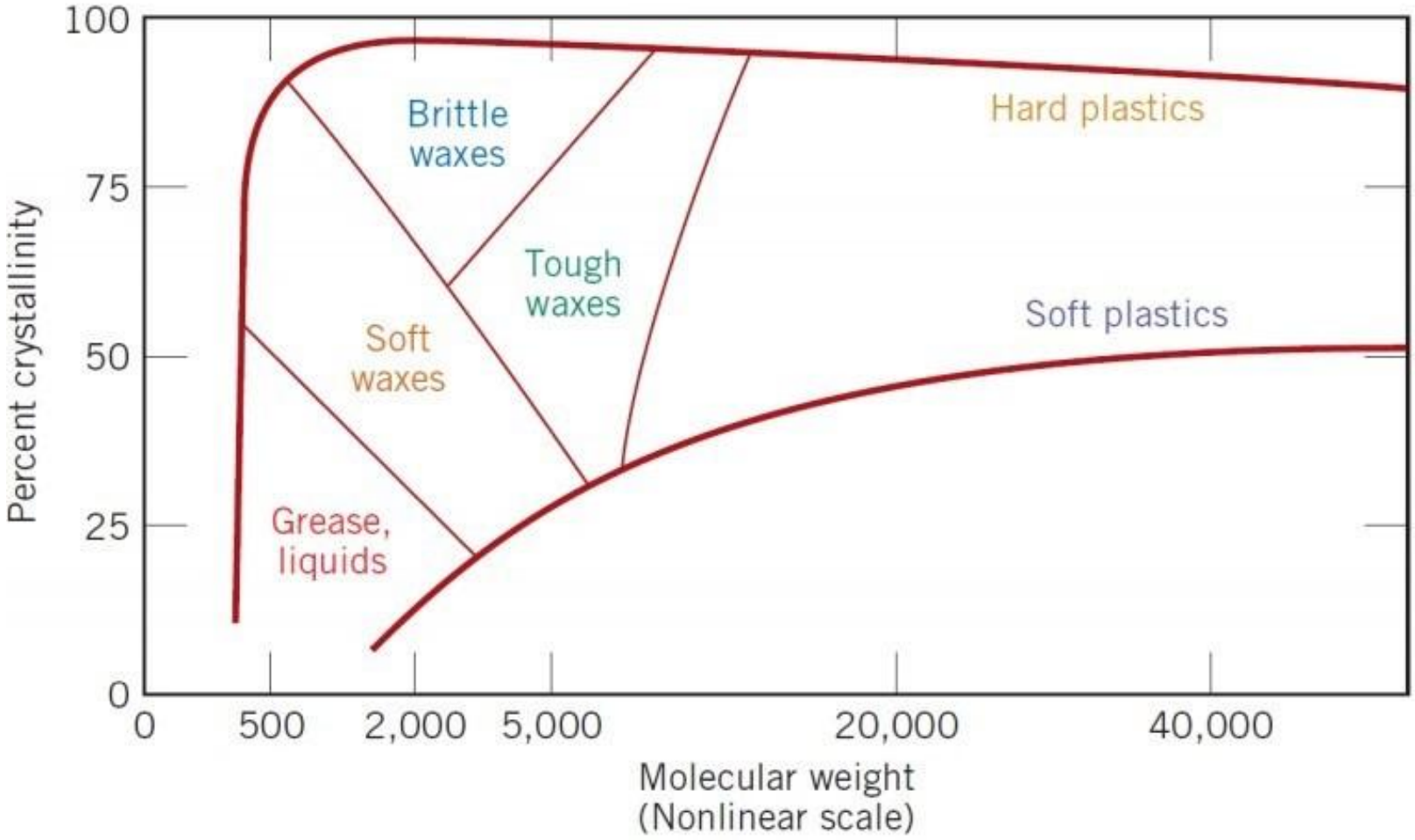
T_g – Glass transition temperature [°K]

X_w – Weight Fraction

Degree of Crystallinity

Degree of crystallinity has a significant influence on the mechanical properties, since it increases the intermolecular secondary bonding and enhances its strength

The influence of crystallinity and molecular weight on the physical properties of polyethylene



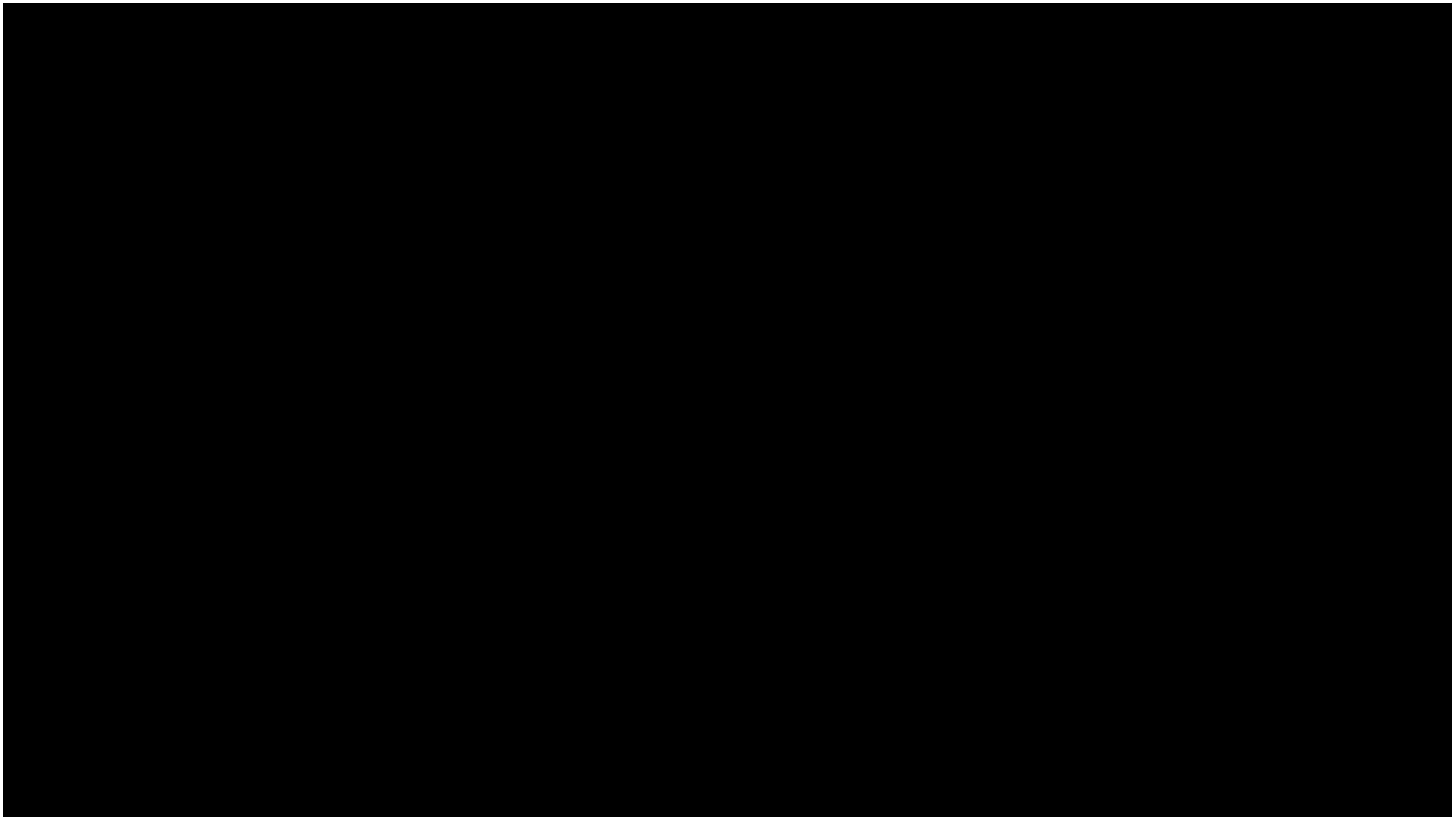
Plastic Polymer Processing

- Fabrication of plastic polymers is usually accomplished by shaping the material in molten form at an elevated temperature.

Common polymer processing techniques include:

- ❑ *Compression*
- ❑ *Injection*
- ❑ *Blow*
- ❑ *Extrusion*
- ❑ *Casting*

Plastic Processing Overview (6 mins)



https://www.youtube.com/watch?v=qn16JtE_vLc